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7 June 1971
Superseding
MIL-G-19027C(WP)
14 March 1966

MILITARY SPECIFICATION

GYRO, STEERING

(For Torpedo Mark 37 Mods 0, 2, 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. Scorn

1.1 Scope. This specification establishes the requirements for the procurement of a steering gyro, NAVORD LD 620060.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of incitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-S-16892	Synchros, 400 cycles
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

DRAWINGS

Naval Ordnance Systems Command

NAVORD LD 620060	Steering Gyro Assembled, and all parts and subassemblies listed thereon

FSC 1355

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NAVORD LD 620062	Shipping (Packing) Condition (Gyro, Steering, Assembled)
NAVORD Drawing 815354	Cover Subassembly
NAVORD Drawing 859711	Cover Subassembly
NAVORD Drawing 877402	Rate Gyro Table Assembled
NAVORD Drawing 696873, 877419, 1623591	Tailcone Schematic Wiring Diagram (Mods 2, 0, and 3, respectively)
NAVORD Drawing 877540	Gyro Test Stand Assembled
NAVORD Drawing 1873928	Steering Gyro Assembled
NAVORD Drawing 2413255	Steering Gyro Schematic Diagram

PUBLICATIONS

Naval Ordnance Systems Command

OS 6341	Ordnance Design Requirements; General, Miscellaneous

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

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3. REQUIREMENTS

3.1 Description. This specification covers a "free gyro," that is, one having three degrees of freedom of rotation, which is used as a course control element in Torpedo Mk 37 Mods 0, 2, and 3. The gyro has an efficiently designed gyro rotor with a high rotational moment of inertia. When this rotor is spinning, it tends to maintain its axis fixed in space regardless of the rolling, pitching, and yawing of the torpedo, or of the rotation of the earth. A Mk 22 synchro transmitter rotor is attached to the gyro outer gimbal and the stator is mounted in the gyro case. Therefore, any turning of the torpedo causes a rotation between the synchro rotor and stator, which generates a voltage that is fed to the control system. The gyro can be remotely set by current Fire Control Systems. A braked servomotor in the gyro turns the synchro stator through a gear train. The brake is normally locked when the gyro is uncaged and the servomotor is in the deenergized condition; the brake is unlocked when the gyro is caged and the servomotor is energized. Therefore, the synchro is used for remotely setting and controlling the torpedo, and the brake prevents the synchro-setting gear train from creeping due to vibration when not in use. The gyro can be caged (held by both gimbals in a fixed position relative to the torpedo) by application of 26 volts dc plus or minus 3 volts. Upon removal of this voltage (at the firing of the torpedo) the gyro is uncaged.

A set of anticircular run (ACR) contacts is incorporated in the gyro so that 165 to 175 degrees of turning of the torpedo in either direction from its firing course will cause the contacts to close, which in turn shuts off the torpedo. Provision is made to withdraw these contacts upon application of 26 volts dc plus or minus 3 volts, which is done when the torpedo enables, thus permitting the torpedo to maneuver freely during its attack.

3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, five steering gyros shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production , that the contractor's production methods are capable of producing steering gyros that comply with the technical requirements of the contract. See 4.3.1.

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3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts-entering into the final assembly of the steering gyro shall conform strictly to the detailed requirements indicated in the drawings and specifications listed in NAVORD LD 620060.

3.2.4 Finishes, coatings, construction and dimensions. Finishes, coatings, construction, and dimensions for the steering gyro shall be strictly in accordance with applicable drawings, specifications, and publications listed in NAVORD LD 620060.

3.2.5 Moisture, fungus, and corrosion. The steering gyro shall be rendered moisture, fungus, and corrosion resistant as required by OS 6341.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in NAVORD LD 620060.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Performance requirements and product characteristics. The steering gyro shall meet the following performance requirements and product characteristics:

3.3.1 Caging. The steering gyro shall have an electrical caging and uncaging arrangement as follows:

3.3.1.1 Caging voltage. The caging voltage, which energizes and unlocks the brake on the servomotor, shall be 26 plus or minus 3 volts dc.

3.3.1.2 Caging contact. The contacts at pins P and R of P201 shall remain open until the gyro is in the full-caged position, at which time the contacts shall be closed.

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3.3.1.3 Caging current. The current required by the caging mechanism during the caging operation shall be 3.4 plus or minus 1.1 amperes, and this current shall drop to 1.4 plus or minus 0.3 amperes while maintaining the gimbals in their caged position.

3.3.1.4 Uncaging the gyro shall be accomplished electrically by removal of the caging voltage described in 3.3.1.1.

3.3.1.5 With the steering gyro stopped or running, the caging mechanism must cage the gyro in 40 seconds maximum time.

3.3.1.6 The steering gyro shall not become uncaged due to acceleration of 20 gravity units (g) in any direction.

3.3.1.7 The steering gyro must operate caged, with the motor running, for a minimum of one hour.

3.3.1.8 When caged and with the gyro motor running, the steering gyro must remain caged when rotated at all speeds up to two revolutions per minute (rpm).

3.3.1.9 When rotating at speeds up to one rpm, the steering gyro shall uncage within 100 milliseconds (ins) after removal of the caging voltage.

3.3.1.10 The gyro gimbal shall not shift more than 1/10 degree when released from the caged to the uncaged position.

3.3.2 Pickoff. The pickoff shall be electrically equivalent to U.S. Navy Synchro Transmitter Mk 22, conforming to the requirements of MIL-S-16892.

3.3.2.1 Zeroing. The synchro shall be so mounted on the gyro assembly that the synchro is electrically zeroed, as described in 4.5.3.6, when the gimbals are in the caged position.

3.3.3 Balance. The gyro shall be balanced and adjusted for operation at zero degrees latitude. Compensation for operation at other latitudes (creep) shall not be incorporated in the gyro assembly.

3.3.3.1 Precession. The allowable precession (drift minus creep due to the earth's rotation), when tested on a roll pitch, and yaw table adjusted to 10 degrees tilt, with the table set for automatic reversing, shall not exceed 0.1 degree per minute average for a 5-minute period, when tested as in 4.5.3.7.

3.3.3.2 Static precession. The allowable static precession (exclusive of creep due to the earth's rotation) shall not exceed 0.15 degree per minute average for a 10-minute period, when tested as in 4.5.3.8.

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3.3.4 Steering gyro motor. The steering gyro motor shall operate with an input of 115 volts plus or minus 5 percent, 400 Hz plus or minus 2 percent, three phase supply.

3.3.4.1 Come-up time. The steering gyro motor shall reach full speed (11,500 to 12,000 revolutions per minute (rpm)) within fifteen seconds. The gyro motor come-up time and speed shall be repeatable within plus or minus two percent.

3.3.4.2 Starting current. The starting current in two of the three power lines (phase A and B) shall not exceed 1.75 amperes. The starting current in the remaining power line (phase C) shall not exceed 1.5 amperes, when tested as in 4.5.3.15.

3.3.4.3 Running current. The running current in two of the three power lines (phase A and B) shall not exceed 0.8 ampere. The running current in the remaining power line (phase C) shall not exceed 0.6 ampere, when tested as in 4.5.3.16.

3.3.5 Anticircular run (ACR) contacts

3.3.5.1 Actuating angle. The steering gyro shall have a set of contacts (CC202 on NAVORD Drawing 2413255) so arranged that between 165 and 175 degrees of gyro rotation in either direction from the caged position shall close the contacts. The difference between the clockwise actuating and counterclockwise actuating angles shall not exceed six degrees.

3.3.5.2 Dwell angle. When actuated as in 3.3.5.1, the contacts shall remain closed for not less than 8 nor more than 18 additional degrees of gyro rotation. The difference between the clockwise and counterclockwise dwell angles shall not exceed six degrees.

3.3.5.3 Provision shall be made to withdraw the contacts described in 3.3.5.1 so they cannot be closed with 360 degrees rotation of the steering gyro.

3.3.5.3.1 The contacts shall be withdrawn upon application of 26 plus or minus 2 volts dc to solenoid SL202, NAVORD Drawing 2413255. At no instant, prior to, during, or after withdrawal of the contacts, shall there be any closed circuit between P or P201 and T of P201. The only exceptions are at the ACR closure points, which occur at 165 to 175 degrees rotation in either direction and with the ACR contacts extended (not withdrawn).

3.3.5.3.2 When the contacts are closed as described in 3.3.5.1, they shall not be damaged by continued rotation of the steering gyro, and the minimum separation of 0.063, specified on Drawing 860101, shall be maintained when the contacts are free from the 165 to 175 degree actuators.

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3.3.6 Remote setting system. The remote setting system must operate compatibly with current Fire Control Systems or their equivalent.

3.3.6.1 Setting accuracy. The gyro must be capable of setting within plus or minus 0.75 degree of any angle set on the fire control system.

3.3.6.2 Slewing speed. The gyro must have a slewing speed of at least 35 degrees per second.

3.3.7 Insulation. All wiring shall withstand a hi-pot test of 500 volts, 60 Hz for one minute between ground and all pins (connected together) of P201, NAVORD Drawing 2413255, and between all mutually insulated pins.

3.3.8 Insulation resistance. The insulation resistance at 500 volts plus or minus 50 volts dc, between ground and all pins (connected together) of P201, NAVORD Drawing 2413255, and between all mutually insulated pins, shall be greater than twenty megohms.

3.3.9 Operating life. The steering gyro shall have an operating life of 500 cycles of 10 minutes per cycle, as. described in 4.4.3.

3.4 Environmental requirements. The steering gyro shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.4.

3.4.1 Vibration. The steering gyro shall meet the following vibration requirements:

3.4.1.1 Vibration, nonoperating. The steering gyro shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.1.2.

3.4.1.2 Vibration, operating. The steering gyro shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.4.1.3.

3.4.2 Temperature and humidity

3.4.2.1 High temperature and humidity, nonoperating. The steering gyro shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.1.

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3.4.2.2 High temperature, operating. The steering gyro shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.2.2.

3.4.2.3 Low temperature, nonoperating. The steering gyro shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.3.

3.4.2.4 Low temperature, operating. The steering gyro shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.2.4.

3.5 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, impregnation of coils, plating, painting, machine screw assemblage, welding, brazing, and freedom Of parts from all burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the steering gyro shall be classified as follows:

(a) Preproduction inspection (paragraph 4.4)

(b) Quality conformance inspection (paragraph 4.5)

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(c) Periodic production inspection (paragraph 4.4)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five steering gyros, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the steering gyros by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract in a lot for production, shipment, or other purpose. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.5, a random sample of 5 steering gyros will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The steering gyros shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.4.1 Vibration

4.4.1.1 Mounting. The steering gyro, less pack or packaging shall be mounted rigidly to a vibration device by means of the four mounting holes shown on Dwg 1873928 and tested as specified in 4.4.1.2 and 4.4.1.3.

4.4.1.2 Vibration, nonoperating. The steering gyro shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.4.1.1.

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4.4.1.3 Vibration, operating. The steering gyro shall be subjected to the "Vibration, operating" test of MIL-E-82590(OS) and during and thereafter shall meet the requirements of 3.4.1.2.

4.4.2 Temperature and humidity

4.4.2.1 High temperature and humidity, nonoperating. The steering gyro shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.4.2.1.

4.4.2.2 High temperature, operating. The steering gyro shall be subjected to the "High temperature, operating" test of MIL-E-82590(OS) and, while at the test temperature, shall meet the requirements of 3.4.2.2.

4.4.2.3 Low temperature, nonoperating. The steering gyro shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.4.2.3.

4.4.2.4 Low temperature, operating. The steering gyro shall be subjected to the "Low temperature, operating" test of MIL-E-82590(OS) and, while at the test temperature, shall meet the requirements of 3.4.2.4.

4.4.3 Operating life. To test for operating life as set forth in 3.3.9:

4.4.3.1 Apply voltages as set forth in 4.5.3.6.2 and 4.5.3.6.3 for a period of 10 minutes.

4.4.3.2 Disconnect the voltages applied in 4.4.3.1 and allow the gyro to coast to a stop.

4.4.3.3 Repeat 4.4.3.1 and 4.4.3.2 through 500 cycles, as specified in 3.3.9.

4.4.3.4 At the conclusion of these tests, the steering gyro shall meet the performance requirements specified in 3.3.

4.4.4 Come-up time. To check come-up time as set forth in 3.3.4.1:

4.4.4.1 Remove the cover, NAVORD Drawing 815354.

4.4.4.2 Apply voltages as set forth in 4.5.3.6.2 and 4.5.3.6.3.

4.4.4.3 Allow the gyro motor to come up to full speed (11,500 to 12,000 rpm), and with a stroboscope tachometer measure and record the revolutions per minute.

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4.4.4.4 Disconnect the voltages and allow the gyro motor to coast to a stop.

4.4.5.5 Again start the gyro motor by applying the voltages prescribed in 4.4.4.2, and check the time it takes for the gyro motor to attain full speed, as measured in 4.4.4.3. The gyro motor come-up time and speed shall be within the limits specified in 3.3.4.1.

4.5 Quality conformance inspection. Each steering gyro delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.1 Test conditions. Unless otherwise specified, the steering gyro shall be subjected to quality conformance tests under the following conditions:

4.5.1.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18.33 degrees C to 35 degrees C).

4.5.1.2 Altitude: Normal ground.

4.5.1.3 Vibration: None.

4.5.1.4 Humidity: Room ambient to 95 percent relative maximum.

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish all necessary test equipment and facilities and shall provide personnel for performing all quality conformance inspections. The test equipment shall be adequate in quantity and, where definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required tests. The following items of test equipment are required to perform the quality conformance tests:

4.5.2.1 Power sources:

4.5.2.1.1 DC power supply, capable of adjustment over the range of 23 to 29 volts, of at least 5 amperes capacity.

4.5.2.1.2 115 volt plus or minus 3 percent, 400 Hertz (Hz) plus or minus 2 percent, 3 phase.

4.5.2.1.3 115 volt, 60 Hz, signal phase.

4.5.2.1.4 500 volt, 60 HZ, single phase.

4.5.2.2 Test stands:

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4.5.2.2.1 Gyro Test Stand Assembled, NAVORD Drawing 877540 or equivalent, with 360 degrees rotation and the ability to set angles to the nearest 1/4 degree.

4.5.2.2.2 Roll, pitch, and yaw table (Navy Stock Number 88-To-798-020-000) 115 volts, 60 Hz, single phase, capable of being set zero to 15 degrees from horizontal with automatic reversing, six cycles per minute with a capacity of 15 pounds.

4.5.2.3 Vacuum tube ac voltmeter, capable of reading 0.1, 0.3, 1, 3, 10, 30, 100, and 300 volts.

4.5.2.4 Matching synchro box. This box shall contain a synchro electrically equivalent to Navy Synchro Control Transformer Mk 14, and shall have provisions for rotating the synchro rotor so that the angle of rotation can be read to the nearest 1/10 of one degree.

4.5.2.5 Rate turntable. Rate Gyro Table Assembled, NAVORD Drawing 877402 or equivalent, operated from 115 volts, 60 Hz single phase with a rotating table capable of turning at speeds up to three rpm.

4.5.2.6 Megohmmeter, capable of measuring insulation resistance at 500 volts dc over the ranges of 0-20-20,000 megohms with an accuracy of 2 percent.

4.5.2.7 One 10k plus or minus 10 percent, 2 watt potentiometer.

4.5.2.8 DC meter, capable of reading current up to 5 amps dc with an accuracy of 3 percent.

4.5.2.9 AC ammeter capable of recording 400 Hz ac, rms current up to 5 amperes, with an accuracy of 3 percent.

4.5.2.10 Ohmmeter.

4.5.2.11 Current Fire Control Systems or their equivalent.

4.5.2.12 Oscilloscope, Tektronix Type 310A or equivalent.

4.5.2.13 Test leads and small tools as required.

4.5.3 Test procedure. The following tests constitute the quality conformance tests of the steering gyro. Unless otherwise specified, all pins and connectors cited are shown on NAVORD Drawing 2413255.

4.5.301 Mount the steering gyro on the rate turntable (4.5.2.5) with the spin axis parallel to the plane of the table.

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4.5.3.2 Connect a jumper across mins M-N of P201 and apply 26 plus or minus 1 volts dc across pins M-P of P201 (pin M being negative). The gyro shall cage as specified in 3.3.1.5. The servomotor brake should be unlocked. Check that it is by manually moving the gear meshed with the servomotor shaft. The gear should move easily.

4.5.3.3 With gyro caged as in 4.5.3.2, apply 115 volts, 400 Hz, 3 phase to pins A-B-C of P201. The gyro must stay caged as specified in 3.3.1.8.

4.5.3.4 With the turntable rotating at a speed up to one rpm, remove the caging voltage (pins M-N and P of P201). The gyro must uncage as specified in 3.3.1.9. The servomotor brake should be locked. Check that it is by attempting to manually move the gear meshed with the servomotor shaft. There should be considerable resistance to moving. (Ref: Holding torque of brake equals 4 oz-in.)

4.5.3.5 Caging current. To check the caging current as set forth in 3.3.1.3:

4.5.3.5.1 Connect the dc ammeter (4.5.2.8) in the 26 volt dc line.

4.5.3.5.2 Cage the gyro by applying 23 volts dc maximum to pins M-P of P201, with pin M negative. The jumper from pin M to N, applied in 4.5.3.2, should be removed prior to making measurements, Reconnect the jumper after this test and leave it connected until 4.5.3.10.1 except during tests of 4.5.3.7 where the jumper is temporarily removed.

4.5.3.5.3 The current shall be 3.4 plus or minus 1.1 amperes while the gyro gimbals are being caged. The current shall drop to 1.4 plus or minus 0.3 amperes when the gyro has caged.

4.5.3.5.4 Using an ohmmeter, monitor continuity between pins P and R of P201, and apply 23 volts dc to pin P(+) and M(-) of P201. There shall be no continuity until the gyro is in the fully caged position, at which time continuity shall exist.

4.5.3.6 Zeroing. To check the electrical zeroing as set forth in 3.3.2.1:

4.5.3.6.1 Mount the gyro with the flat cover (NAVORD Drawing 859711) down on the gyro test stand (4.5.2.2.1).

4.5.3.6.2 Cage the gyro by applying 26 volts dc to pins M-P of P201, with pin M negative.

4.5.3.6.3 Start the gyro by applying 115 volts, 400 Hz, 3 phase to pins A-B-C of P201 (phase A to pin A, phase B to pin B, phase C to pin C).

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4.5.3.6.4 Connect a jumper between pins B-K of P201.

4.5.3.6.5 Measure the ac voltage across pins A-J of P201. The voltage shall measure more than 160 volts. This is approximately zero setting.

NOTE

In order to obtain these measurements, it may be necessary to change the synchro position by running motor M202 electrically.

4.5.3.6.6 Remove jumper between pins B-K of P201, and measure ac voltage across pins J-L of P201. This voltage should be less than 0.3 volt.

4.5.3.6.7 Remove the voltages applied in 4.5.3.6.2 and 4.5.3.6.3.

4.5.3.6.8 Connect the gyro as shown in Figure 1.

4.5.3.6.9 Adjust the 10k pot so that the ac voltmeter idicates 15 plus or minus 2 volts.

4.5.3.6.10 Uncage the gyro (remove power source from pins M-P of P201), and manually turn the gyro in a counterclockwise direction (5 to 10 degrees) as viewed from above. This shall increase the voltmeter reading.

4.5.3.6.11 This completes the test for electrical zeroing. Disconnect all power sources and remove all connections shown in Figure 1. Allow the gyro to cool to ambient before proceeding on the next tests.

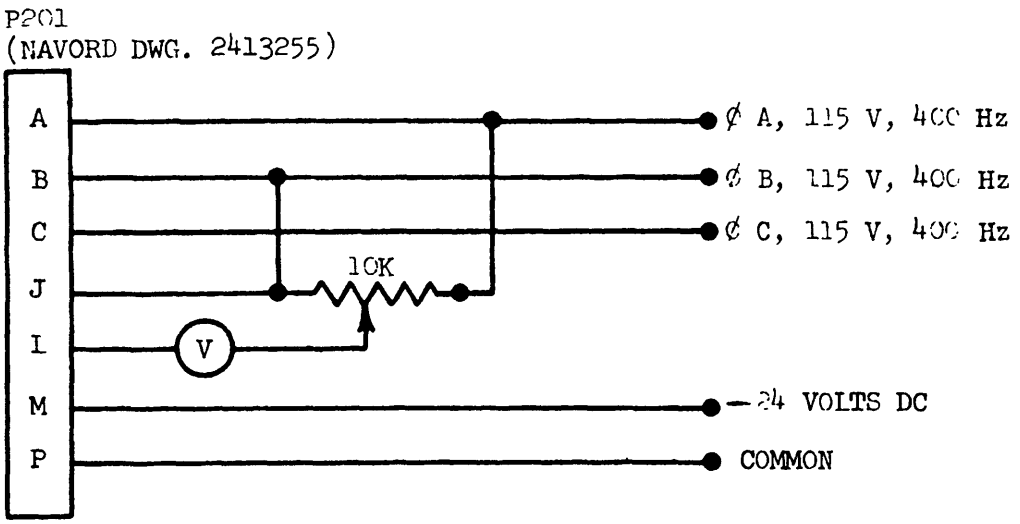


FIGURE 1

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4.5.3.7 Precession. To determine the gyro precession as described in 3.3.3.1:

NOTE

Prior to start of drift checks, the gyro shall have had a total run time of from 5 to 10 minutes from the ambient point specified in 4.5.3.6.11.

4.5.3.7.1 Connect the gyro, as shown in Figure 2, and mount it on the roll, pitch, and yaw table (4.5.2.2.2) with the connector facing north.

4.5.3.7.2 Set the matching synchro box (4.5.2.4) so that the ac voltmeter indicates minimum voltage. This voltage shall be less than 0.3 volt ac.

4.5.3.7.3 Uncage the gyro by removing voltage from pins M-P of P201, and immediately readjust the synchro box dial as specified in 4.5.3.7.2.

4.5.3.7.4 The uncaging error, the difference between synchro box angular indication in 4.5.3.7.2 and 4.5.3.7.3, shall be as specified in 3.3.1.9.

4.5.3.7.5 Set the synchro box as specified in 4.5.3.7.2 and record the angular indication on the synchro box.

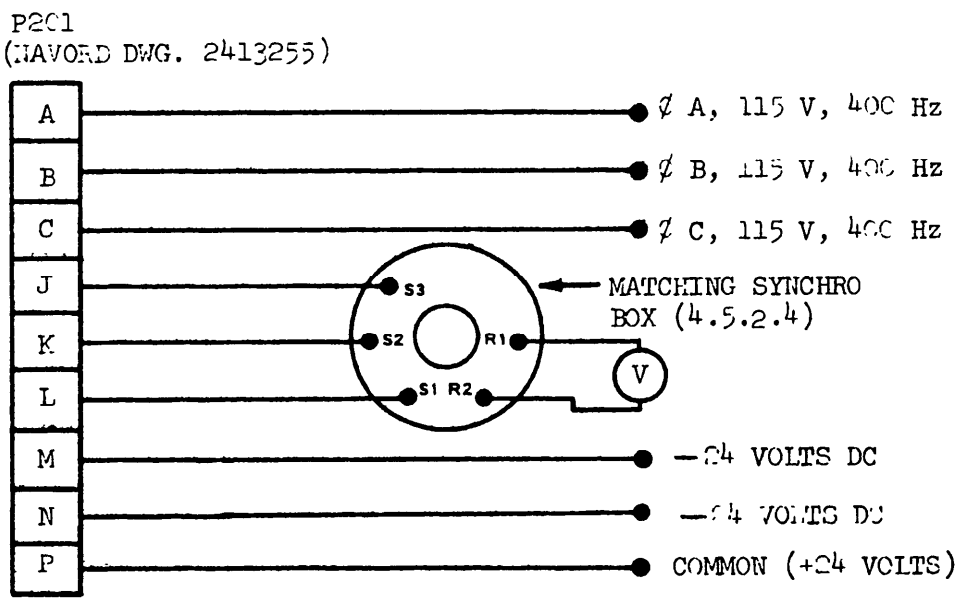


FIGURE 2

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4.5.3.7.6 Start the roll, pitch, and yaw table tilted 10 degrees, with the table set for automatic reversing. After five minutes, stop the table, return it to a level position, and again set the synchro box as specified in 4.5.3.7.2.

4.5.3.7.7 Record the synchro box angle reading and subtract it from the reading of 4.5.3.7.5 to obtain the drift.

4.5.3.7.8 Repeat 4.5.3.7.5 through 4.5.3.7.7 with the open end of the gyro rotor-facing east, south, and west. The gyro is placed in the above orientations by caging the gyro facing the connector east, south, and west, respectively, uncaging the gyro, and refacing the connector north. (Note: Advance the matching synchro 90 degrees for each successive reading and withdraw the ACR contacts for south and west readings.)

4.5.3.7.9 Creep, in degrees per minute, is 0.25 times the sine of the latitude; that is, if the latitude is 43 degrees and 50 minutes, the apparent drift is 0.17 degrees per minute.

4.5.3.7.10 Recession, in degrees per minute, is equal to the total drift for a given period corrected for creep, divided by the time in minutes for that period. Precession shall not exceed plus or minus 0.1 degree per minute.

4.5.3.7.11 As precession can be opposite to creep, the following procedure shall be used to ensure that consecutive drift readings are properly corrected for the effect of creep.

4.5.3.7.11.1 When setting the synchro box in 4.5.3.7.6 and 4.5.3.7.8, observe the direction in which the synchro is turned to null out the error. If rotation is counterclockwise, the difference between the second synchro reading and the first (observed drift) shall be considered “positive.” If the rotation is clockwise, the difference shall be considered “negative.”

4.5.3.7.11.2 As creep in the northern hemisphere is clockwise, its value as calculated by 4.5.3.7.9 shall be considered “positive.” Creep in the southern hemisphere is counterclockwise and shall be considered “negative.”

4.5.3.7.11.3 Total precession for a given period is equal to the algebraic difference between the observed drift for that period and the calculated creep for the same period.

4.5.3.7.11.4 This completes this portion of the drift check.

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4.5.3 .7.11.5 Turn the gyro manually less than one degree in a clockwise direction, as viewed from above.

4.5.3.7.11.6 In the northern hemisphere, if the voltage obtained from 4.5.3.7.11.5 decreases, then the creep is subtracted from the total drift to obtain the error drift.

4.5.3.7.11.7 If the voltage obtained in 4.5.3.7.11.5 increases instead of decreases, the gyro is drifting faster than the creep due to the earth's rotation, and the creep must then be added to the total drift to obtain the error drifts.

4.5.3.7.11.8 This completes this portion of the drift check. Disconnect all power sources and remove all connections as shown in Figure 2. Know the gyro to cool to ambient.

4.5.3.8 Static precession. To measure the static precession as described in 3.3.3.2:

4.5.3.8.1 Mount the gyro on the gyro test stand.

4.5.3.8.2 Connect the 10k potentiometer (4.5.2.7) across pins J and L of P201 as a load, and connect the ac voltmeter (4.5.2.3) across this 10k load also.

4.5.3.8.3 Cage the gyro by applying 26 volts dc to pins M-P of P201, with M negative. Apply 115 volts, 400 Hz, 3 phase to pins A-B-C of P201.

4.5.3.8.4 Allows 60 second minimum warmup time, then uncage the gyro and rotate the gyro until a null is reached on the ac voltmeter.

4.5.3.8.5 Allow the gyro to run 10 minutes or longer uncaged. At the end of that time rotate the gyro in the direction to obtain a null on the ac voltmeter. The time of the run and the number of degrees the gyro was turned to obtain the second null should be recorded. From these the total drift in degrees per minute can be found.

4.5.3.8.6 The creep calculated in 4.5.3.7.9 shall be added or subtracted from the total drift, as determined in 4.5.3.7.11.

4.5.3.8.7 The static precession shall be as specified in 3.3.3.2.

4.5.3.8.8 Remove connections made in 4.5.3.8.2.

4.5.3.9 Anticircular run contacts. To check the ACR contacts, as set forth in 3.3.5:

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4.5.3.9.1 Apply voltages as set forth in 4.5.3.6.2 and 4.5.3.6.3. There shall be an open circuit between pins P-T of P201.

4.5.3.9.2 Remove the caging voltage as set forth in 4.5.3.6.2 (pins M-P of P201). Manually rotate the gyro in a clockwise direction, as viewed from above. At any angle up to 165 degrees, pins P-T of P201 must show an open circuit. At some angle between 165 and 175 degrees, the circuit between P-T must close. Record this angle.

4.5.3.9.2.1 Continue to rotate the gyro in a clockwise direction until the circuit between P-T of P201 opens. Record this angle.

4.5.3.9.2.2 The difference between the angles recorded in 4.5.3.9.2 and 4.5.3.9.2.1 shall be between 8 and 18 degrees.

4.5.3.9.3 Again apply voltage as set forth in 4.5.3.6.2.

4.5.3.9.4 Remove the caging voltage as set forth in 4.5.3.6.2 and manually rotate the gyro in a counterclockwise direction, as viewed from above. At any angle up to 165 degrees pins P-T of P201 must show an open circuit. At some angle between 165 and 175, the circuit between P-T must close Record this angle.

4.5.3.9.4.1 Continue to rotate the gyro in a counterclockwise direction until the circuit between P-T of P201 opens. Record this angle.

4.5.3.9.4.2 The difference between the angles recorded in 4.5.3.9.4 and 4.5.3.9.4.1 shall be between 8 and 18 degrees.

4.5.3.9.4.3 The difference between the actuating angles of 4.5.3.9.2 and 4.5.3.9.4 shall not exceed six degrees.

4.5.3.9.4.4 The difference between the dwell angles calculated in 4.5.3.9.2.2 and 4.5.3.9.4.2 shall not exceed six degrees.

4.5.3.10 To withdraw the ACR contacts, as set forth in 3.3.5.3:

4.5.3.10.1 Remove jumper between pins M-N of P201, applied in 4.5.3.2.

4.5.3.10.2 Connect one end of a 1000 ohm, 1 watt resistor to pin T of P201, and apply 26 volts dc between the other end of the resistor and pin P of P201. Connect the vertical plates of an oscilloscope across the resistor. Apply 26 volts dc, specified in 4.5.2.1.1, to pins P-N of P201 (pin N is negative). Observe that no voltage pulse occurs just prior to, during, or immediately after application of voltage to pins P-N of P201. After application of voltage to P-N, rotate gyro through 360 degrees. The circuit between pins P-T must remain open.

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4.5.3.11 To check if the ACR contacts will be damaged by continued rotation, as set forth in 3.3.5.3.2:

4.5.3.11.1 Connect jumper between pins n-N of P201.

4.5.3.11.2 Apply voltages as set forth in 4.5.3.6.2 and 4.5.3.6.3. The gyro shall cage as specified in 3.3.1.4.

4.5.3.11.3 Remove caging voltage from pins M-P of F201.

4.5.3.11.4 Manually rotate the gyro 360 degrees, as viewed from shove.

4.5.3.11.5 Repeat the tests set forth in 4.5.3.9 to ensure that no damage resulted to the ACR contacts after the gyro was rotated as set forth in 4.5.3.11.4.

4.5.3.12 Remote setting system. To test the remote control setting system as described in 3.3.6:

4.5.3.12.1 Mount the gyro on the calibrated gyro test stand (4.5.2.2.1) with the flat base (NAVORD Drawing 859711) down.

4.5.3.12.2 Set the gyro test stand to zero degrees.

4.5.3.12.3 Connect the gyro to current Fire Control Systems or their equivalent, as in Figure 3.

4.5.3.12.4 Cage the gyro according to instructions on the fire control system.

4.5.3.12.5 Set the gyro dial on the fire control system to 90 degrees port, and actuate the setting control until the indicating device (tuning eye) shows the synchros to be aligned.

4.5.3.12.6 Place the ac vacuum tube voltmeter (4.5.2.3) across pins J-L of P201.

4.5.3.12.7 Uncage the gyro according to instructions on current Fire Control Systems.

4.5.3.12.8 Turn the gyro in a counterclockwise direction, as viewed from above, until the voltmeter indicates null voltage (0.3 volt or less).

4.5.3.12.9 The time elapsed from the instant the gyro is uncaged (4.5.3.12.7) until the gyro has been turned to its minimum voltage position (null voltage 4.5.3.12.8) must be less than 30 seconds to prevent the total gyro drift from affecting the test reading in 4.5.3.12.10 and 4.5.3.12.15.

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4.5.3 .12.10 Read the angle on the gyro test stand, as obtained in 4.5.3.12.8. This angle shall be 90 plus or minus 0.75 degrees counter-clockwise.

4.5.3.12.11 Reset the gyro test stand to zero degrees.

4.5.3.12.12 Cage the gyro as in 4.5.3.12.4.

4.5.3.12.13 Set the gyro dial on the fire control system to 90 degrees starboard, and actuate the setting control until the indicating device (tuning eye) shows the synchros to be aligned.

4.5.3.12.14 Repeat 4.5.3.12.6 and 4.5.3.12.7.

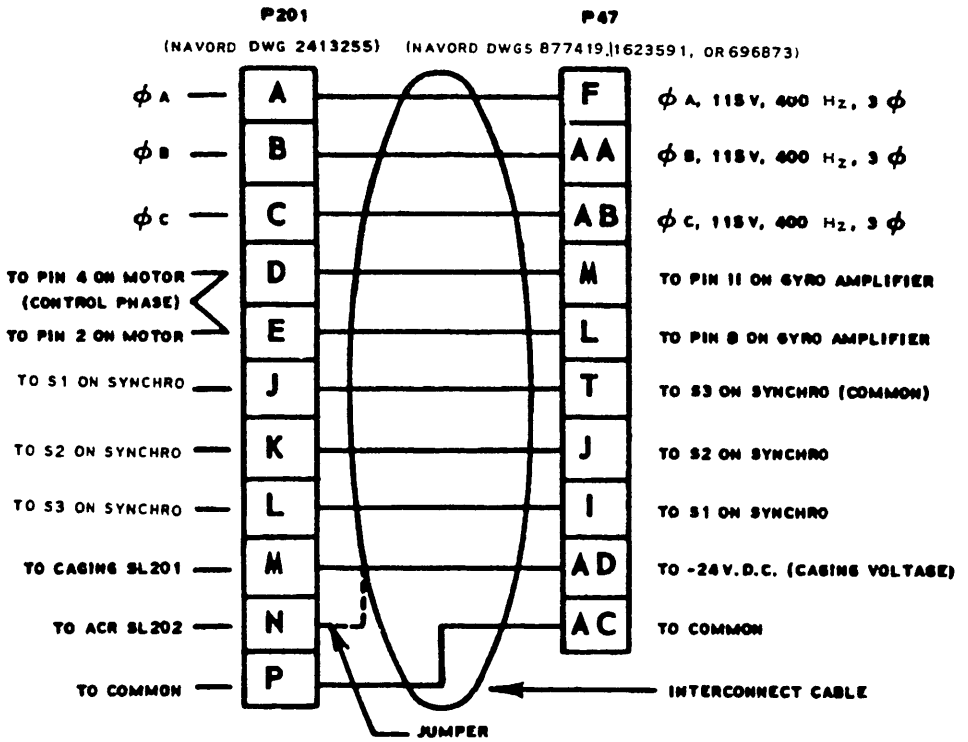


FIGURE 3

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4.5.3.12.15 Turn the gyro in a clockwise direction, as viewed from -above, until the voltmeter indicates null reading (0.3 volt or less). Observe the time limitations specified in 4.5.3.12.9.

4.5.3.12.16 Read the angle on the gyro test stand as obtained in 405.3.12.15. This angle shall be 90 plus or minus 0.75 degrees clockwise.

4.5.3.12.17 Reset the gyro test stand to zero degrees.

4.5.3.12.18 Cage the gyro as in 4.5.3.12.4.

4.5.3.12.19 Set the fire control angle on Fire Control System to zero degrees and actuate the setting control until the indicating device (tuning eye) shows the synchros to be aligned.

4.5.3.12.20 Set the gyro dial on the fire control system to 180 degrees in either direction (port or starboard), and actuate the setting control until the indicated setting as shown by the tuning eye shows the sychros to be aligned.

4.5.3.12.21 Measure the time from the actuation of the setting control until the synchros are aligned (4.5.3.12.20). The time shall be less than five seconds. This test is the measure of slewing speed.

4.5.3.13 Insulation. To test the insulation between pins and case, as specified in 3.3.7, apply 500 volts, 60 Hz for one minute between all pins of P201, connected together, and ground. There shall be no evidence of insulation breakdown or arcing.

4.5.3.13.1 To test insulation between pins, as specified in 3.3.7, apply 500 volts, 60 Hz for one minute between all mutually insulated pins of P201. There shall be no evidence of insulation breakdown or arcing.

4.5.3.14 Insulation resistance. To measure the insulation resistance between all pins and case, as specified in 3.3.8, connect the megohmmeter (4.5.2.6) across all pins of P201, connected together, and ground. The resistance shall be greater than 20 megohms.

4.5.3.14.1 To measure the insulation resistance between all mutually insulated pins, as specified in 3.3.8, connect a megohmmeter (4.5.2.6) between all mutually insulated pins of P201. The resistance shall be greater than 20 megohms.

4.5.3.15 Starting current. With the gyro connected as specified in 4.5.3.6.3 and with the 400 Hz ammeter (4.5.2.9) connected in each of the

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three lines (pins A-B-C of P201), measure the 400 Hz current when the power is first applied. The starting current shall be as specified in 3.3.4.2.

4.5.3.16 Running current. With the gyro connected and running as specified in 4.5.3.6.3, allow the motor to gain speed for 15 seconds and again measure the 400 Hz current in each of the three lines (pins A-B-C of P201). The running current shall be as specified in 3.3.4.3.

4.5.3.17 Visual and dimensional examination. Each steering gyro shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.5.3.18 Packaging, packing, and marking. Packaging, packing, and markings shall comply with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The gyro shall be packaged in accordance with Method IID of MIL-P-116 and LD 620062.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The gyro shall be packaged in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Number of packages per pack. The number of unit packages per pack, for Level C, shall be as specified in the contract or order.

5.2.2 Level A

5.2.2.1 Unit pack. The gyro shall be packed in accordance with LD 620062 and may be shipped individually in small lots.

5.2.2.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Load Type 111 (see 6.2).

5.2.3 Level B. Not applicable.

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5.2.4 Level C. The gyro shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal markings, Level A. Unit packs shall be marked in accordance with LD 620062.

5.3.2 Normal markings, Level C. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.3 Serial numbers. Unit packages and shipping containers shall be marked with the serial numbers of the units contained within.

6. NOTES

6.1 Intended use. The gyro is intended for use in Torpedo Mk 37 Mods 0, 2, and 3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production tests.

(f) Data required other than that specified in 3.2.1.

(g) That all additional samples required shall be provided and inspected at the expense of the supplier when such additional samples are required because of failure of any unit(s) to pass prescribed periodic production inspections.

6.3 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

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Level C. This level shall be used only when it is definitely known that the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

Preparing Activity
Navy - OS
Project No. 1355-N059

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